

Agricultural Monitoring of Beehive Health

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Abstract

Bee populations are on decreasing with the advent of a scenario known as Colony Collapse Disorder(CCD). Bees are vital pollinators and play a major role in our food supply. Bee mortality rates have been heightened to around 30% in the recent years and the reasons are still not confirmed. With the advancement of technology and Internet of All Things devices, automation could be used to facilitate the monitoring of hives. This research uses computer vision and audio analysis in order to monitor the beehive health. Our computer vision algorithm has an accuracy rate of 91% and is used to determine population metrics of the hive. Audio frequency analysis is used to determine orientation flights and differentiate between different type of hives. This information is used to supplement the computer vision. After passing through the local outlier factor algorithm, the state of the beehive will be live updating to the intended caretaker and send alerts for any emergencies.

Keywords: colony collapse disorder, detecting bee populations, monitoring beehives

1. Introduction

The annual beehive mortality rates have been increasing from 10% to 30% leading to an occurrence known as colony collapse disorder. Bees are vital to our ecosystem and are one of the leading pollinators making sure we have a healthy and stable food supply. Bees are the backbone of our agricultural sector and this dramatic reduction in population is alarming. Therefore, it is important to determine a method to monitor beehives and notify caretakers of any disturbances or signs of colony collapse. This research endeavors to implement an innovative way of monitoring beehive health that could help prevent the occurrence of colony collapse disorder caused by a significant decline of bee populations. Our method involves the use of audio and computer vision to determine the state of the beehive.

2. Literature Review

Bees are extremely vital to our ecosystem and colony collapse disorder poses many detrimental effects on the food supply to come. Research from the United States Department of Agriculture states 80% of our diet that comes from pollinated crops are from bees. There are many crops which depend on bees, and they contribute over 24 billion dollars to the U.S. economy. Bees are extremely vital. The causes of colony collapse disorder are still being researched but some probable causes include the increase of the varroa mite, diseases and exposure to pesticides. The massive decline in populations has devastated many farmers.

Previous research focusing on beehives has focused on acoustic methods for swarming as well as fuzzy inference systems in order to determine CCD. Others have used weight, temperature and humidity to assist with their predictions. Some previous researchers have also implemented

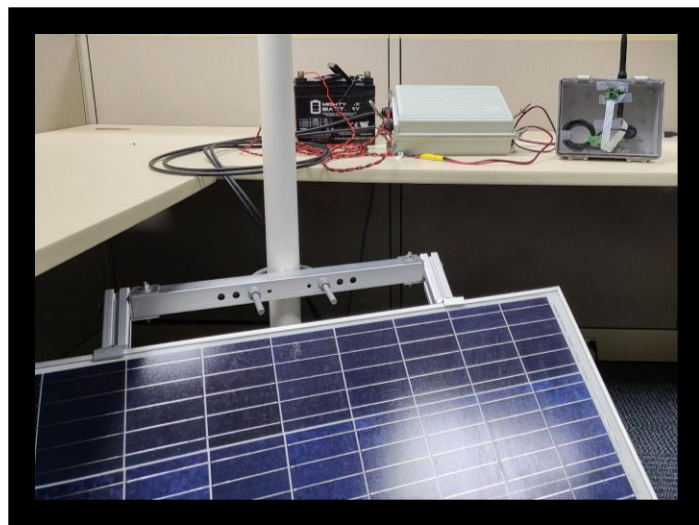
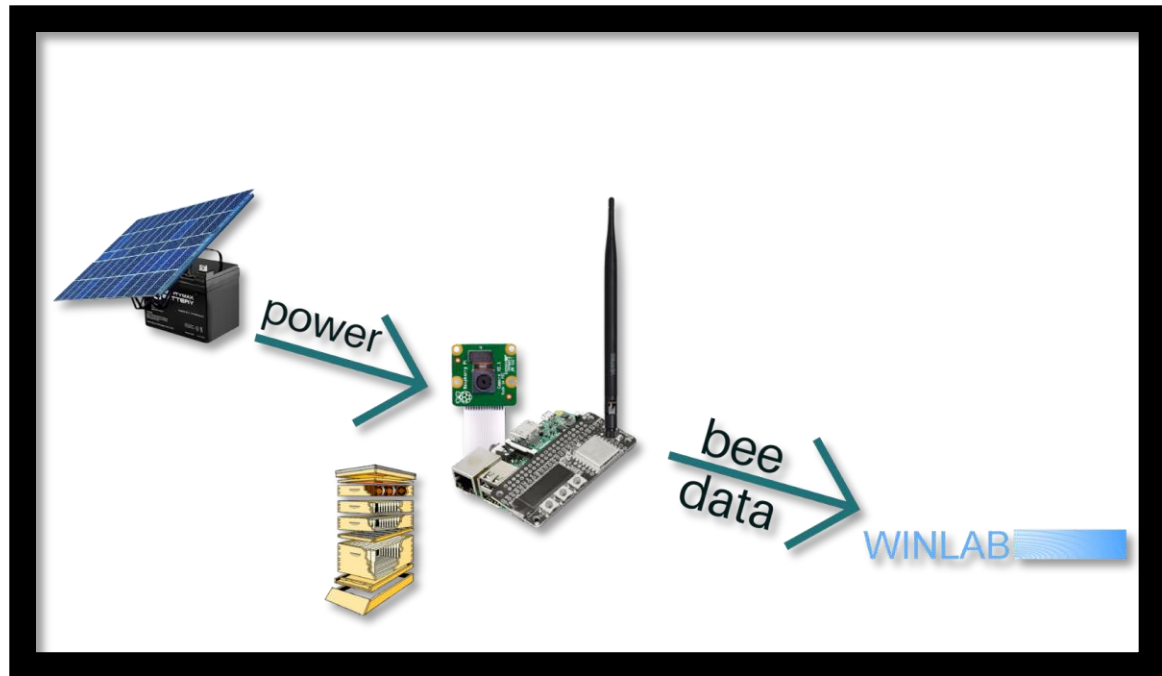
computer vision techniques detecting varroa mites among other invaders which are probable reasons for CCD.

3. Experimental Design

The research conducted aimed to analyze the state on the Beehive using computer vision and audio analysis. A system was created that was powered by battery which uses solar power as its energy source. The system includes a camera, raspberry pi, antenna as well as a Long-Ranged Radio(LORA) to monitor the Beehives in real-time. This data is uploaded to our firebase online data server where a series of algorithms are used to determine if there is any emergency in the hive. If there are any problems, an alert is sent using Twilio to the beekeeper's phone or email.

3.1 System Setup

The solar panel used to power the entire system was a 100W solar panel which supported a 12V battery and had a power output with a 12V socket. When measuring the power usage from our raspberry pi, we noticed that only 0.5 A of current from the power supply was used, which is well below what is sustainable with a solar panel. But because some days might be cloudy, and with a few days of minimal sunshine the system might not have enough power, we decided that a raspberry pi was the best option to us compared to more robust systems. The solar panel powers the raspberry pi, which is fitted with a LORA, antenna, microphone as well as camera. Installed at the Rutgers Horticulture farm, the raspberry pi sends data from the hive back to Rutgers's Winlab using the radios. Here the data is uploaded to a firebase database, where live monitoring results are displayed, from here our local outlier factor algorithm is used to help determine the state of the Beehive.



3.2 Long-Ranged Radio(LORA)

The beehive is located at the Rutgers University horticulture farm which is around 1-2 miles away from laboratory. There are not Wi-Fi capabilities at the system placing some limitation on sending data from the farm to the laboratory. To evade this measure a few methods were explored including Bluetooth, but its range does not reach the desired distance to send data back and forth from the farm and the lab. So, LORA(Long-Ranged Radio) was determined as the best possible

solution. Capitalizing on the long range of LORA, data communication is possible, but a limitation is the low bandwidth possible. To counter this measure, since importing real time videos are not possible, data must be simplified to initiate data transfer. Data is sent through the form of UDP packets to upload to the cloud and perform analysis to monitor the beehive.



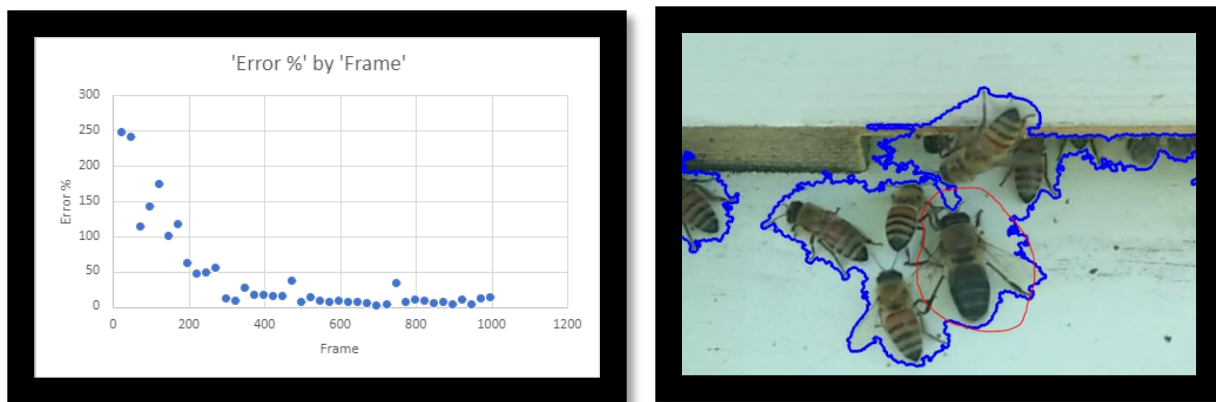
4. Results

4.1 Computer Vision

To determine the state of the beehive, it is important to measure the population of the hive. Determining the total count of the hive is improbable with just a camera due to that fact that bees will be stacked on top of each other and getting an accurate count will be near impossible. But placing a camera near the entrance of the hive, we can approximate the metrics and determine steep decline of bee populations. Initially, we created a YOLOv4 model through the darknet framework. YOLOv4 is one of the most accurate convolutional neural networks based deep learning models used for object detection and image classification. Feeding into the algorithm over 9000 images, an accuracy rate of approximately 90% was obtained. However, as we realized, running this algorithm was extremely power intensive and a raspberry pi would not be able to handle such an algorithm. Therefore we had to sacrifice processing power for accuracy, and we explored two different strategies: background subtraction and hue color classification.

4.1.1 *Background Subtraction*

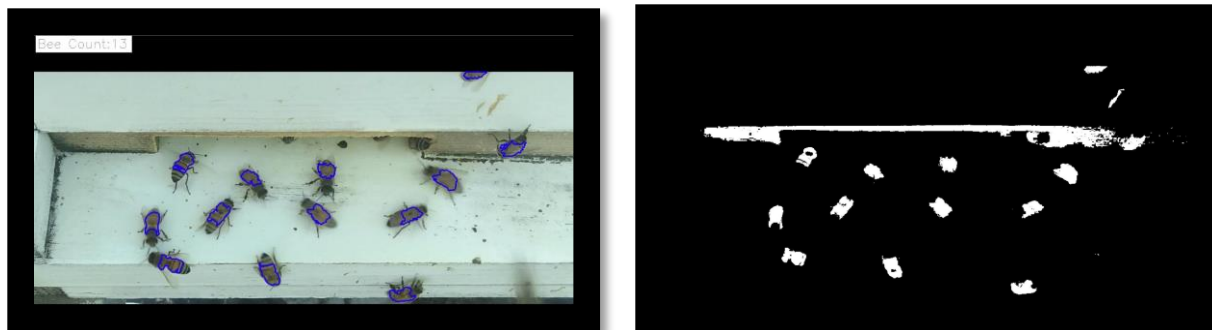
The background subtraction algorithm detects moving objects within the camera frame which in this case detects bees. A K-Nearest-Neighbors Background Subtraction Algorithm was implemented with a history of 500 frame, distance threshold of 400 and shadows as false. In order to limit noise contour areas less than 1100 were not counted. In addition Gaussian blur image processing techniques were used to increase accuracy of the detection. A major observation from this data was that contour accuracy increased as frame rate increased. For the entire sample the accuracy rate was 60.4% while for the last 500 frames the accuracy rate was 92.64%. The accuracy of the model increased because the KNN algorithm detects moving objects by coming each frame to its previous frames, so as the frame count increased the accuracy increased as well. But there were a few errors with this algorithm, it would not differentiate between bees with different sizes, for example drone bees were counted as 2 bees because the method only accounted for workers bees which were smaller in size but more plentiful among the hive. Also the model did not account for stagnant bees that were not moving, some bees were at the entrance of the hive and were stagnant for long periods of time, probably to regulate temperature flow withing the hive.



4.1.2 *Hue Color Classification*

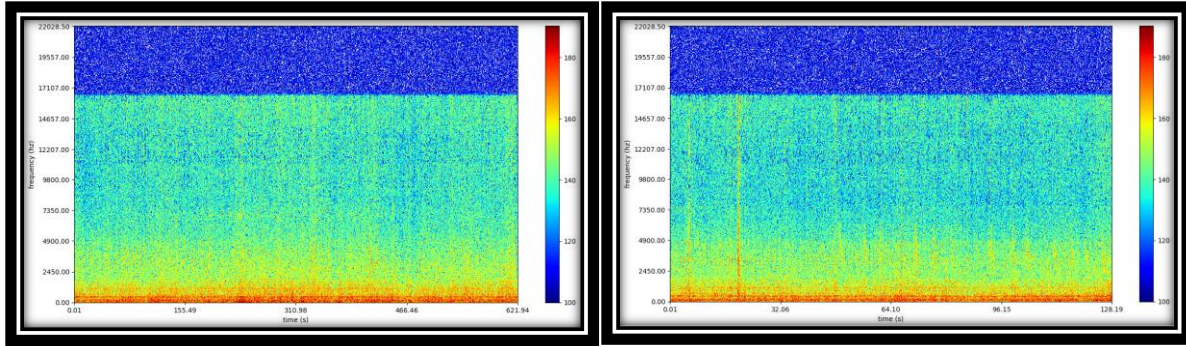
To limit these errors we decided to work with color Hue based classification instead. By using the color of the pixels to choose the color spectrum ranges of the bee, bees were able to be

detected in the videos. Instead of RGB, since HSV is more reliable than RGB for computer vision algorithm, we decided to move forward with HSV and extract the hue of the bees only. The accuracy rate for this new classification was a whopping 91% which was greater than our YOLOv4 model itself. For the most part the algorithm worked perfectly, but the errors primarily resulted from camera positioning, videography, lighting, and the bee shadows in the video.



4.2 Audio Analysis

Audio analysis was used as an external tool to supplement the computer vision techniques. Initially, amplitude analysis was experimented with as our method, which determines how loud the beehive is to count the number of bees. By producing a spectrogram of the audio signal, and isolate certain frequencies to separate background noise from bees an audio spectrum was generated. The problem with this technique is that the algorithm that it will incorrectly classify as a large number of bees if there is only one but is close to the microphone compared to many that are away. So instead of using amplitude analysis, frequency pitch analysis was used. Because there wasn't enough data to compare health vs unhealthy hives, we decided to use frequencies to determine when bees are doing orientation flights. This will determine the number of new bees in the hive which can be used to determine the state of the hive as well. A high frequency filter was also used to remove background noise in the hive. By comparing the sounds of a healthy hive to those of an unhealthy hive, we can predict and detect colony collapse disorder.



4.3 Local Outlier Factor Algorithm

The local outlier factor algorithm is used to find outliers in data which can be used to detect and remove anomalies in data to make our data samples more accurate to the beehive. This machine learning algorithm uses information regarding the local deviation of a point in relationship to its neighbors to determine outliers.

Conclusion

This research project culminated with a few key takeaways. The bee counting algorithm shows accuracy of around 91%. The Raspberry Pi receives a sufficient 0.5 A of current. The system is weatherproofed for outdoor implementation. One problem we faced was that the radio sends data over the distance of about half a mile rather than theoretical two miles. A potential solution to this problem is to use a third radio as a relay node to increase range of sending data. Furthermore, a potential avenue to work on for the future is to implement temperature/humidity detection as an additional way of monitoring beehive population.

5. References

- “Colony Collapse Disorder and Pollinator Health.” *Colony Collapse Disorder and Pollinator Health* | *National Invasive Species Information Center*,
<https://www.invasivespeciesinfo.gov/subject/colony-collapse-disorder>.
- Ferrari, S., et al. “Monitoring of Swarming Sounds in Bee Hives for Early Detection of the Swarming Period.” *Computers and Electronics in Agriculture*, vol. 64, no. 1, 2008, pp. 72–77.,
<https://doi.org/10.1016/j.compag.2008.05.010>.
- Kulyukin, Vladimir, et al. “Integration of Scales and Cameras in Nondisruptive Electronic Beehive Monitoring: On the within-Day Relationship of Hive Weight and Traffic in Honeybee (*Apis Mellifera*) Colonies in Langstroth Hives in Tucson, Arizona, USA.” *Sensors*, vol. 22, no. 13, 2022, p. 4824., <https://doi.org/10.3390/s22134824>.
- Kulyukin, Vladimir, et al. “Toward Audio Beehive Monitoring: Deep Learning vs. Standard Machine Learning in Classifying Beehive Audio Samples.” *Applied Sciences*, vol. 8, no. 9, 2018, p. 1573.,
<https://doi.org/10.3390/app8091573>.
- Kviesis, Armands, et al. “Application of Fuzzy Logic for Honey Bee Colony State Detection Based on Temperature Data.” *Biosystems Engineering*, vol. 193, 2020, pp. 90–100.,
<https://doi.org/10.1016/j.biosystemseng.2020.02.010>.
- Voudiotis, George, et al. “Deep Learning Beehive Monitoring System for Early Detection of the Varroa Mite.” *Signals*, vol. 3, no. 3, 2022, pp. 506–523.,
<https://doi.org/10.3390/signals3030030>.

Williams, Geoffrey R., et al. "Colony Collapse Disorder in Context." *BioEssays*, vol. 32, no. 10, 2010, pp. 845–846., <https://doi.org/10.1002/bies.201000075>.